Problem 1

(a) The female and non-binary group is disadvantaged by this way of imputation. The value used for imputation is the overall mean years of experience (6.12), which is higher than the mean years of experience of men (5.6-5.7) but lower than the mean year of experience of female and non-binary applicants (7.4-7.9). Since the model prefers applicants with more experience, female and non-binary applicants with missing values are misrepresented. They are ranked lower than they should be.

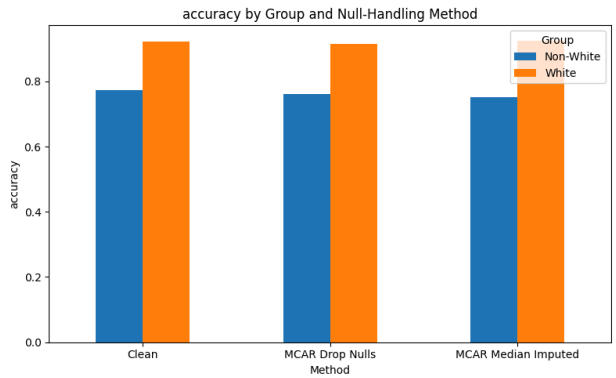
(b) The pre-existing bias is basically that the data already shows differences in experience across groups. Female and non-binary applicants have more experience than male applicants. These differences may come from structural inequalities in the real world. The way of imputation in (a) underestimates the experience of female and non-binary applicants. It amplifies the existing bias.

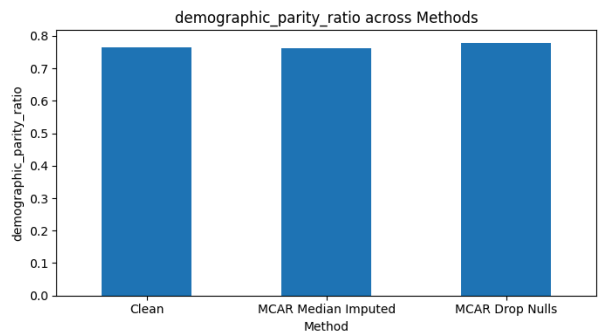
(c) I will use group-wise mean imputation. It is slightly different from the method used in part (a). This method fills in the missing values within each sex-race group rather than the overall mean. This method preserves the group-specific experience distributions. I think this approach aligns with fairness through awareness: individuals are treated relative to their group context.

Problem 2

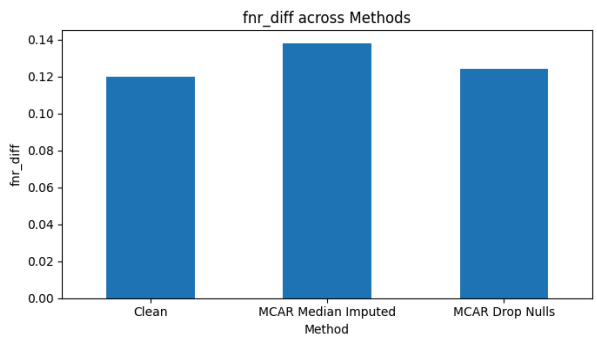
(a) Equal opportunity is most appropriate because it ensures that equally qualified students have equal chances of being selected across racial groups.

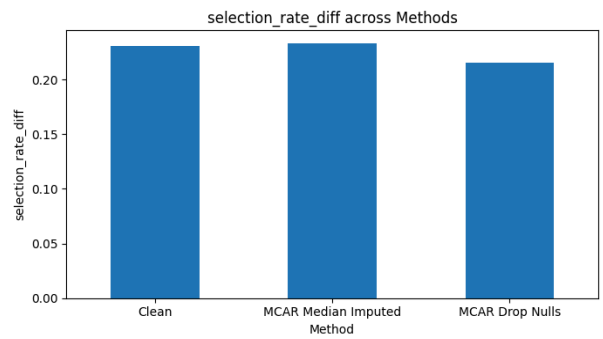
(b) Under the MCAR assumption, missingness is independent of both the features and the sensitive attribute (race). Therefore, we expect that neither deletion nor median imputation will introduce systematic bias across groups. However, both methods may have worse overall performance due to information loss or distortion. Dropping columns reduces the sample size and may cause higher variance, while median imputation preserves the sample size.





From the results, we can see that compared to clean models, both MCAR methods show similar group disparities. There is no dramatic shift in fairness metrics. The FNR difference slightly increases under median imputation, suggesting a small increase in unequal treatment of qualified individuals. The demographic parity ratio and selection rate difference remain relatively stable across methods.



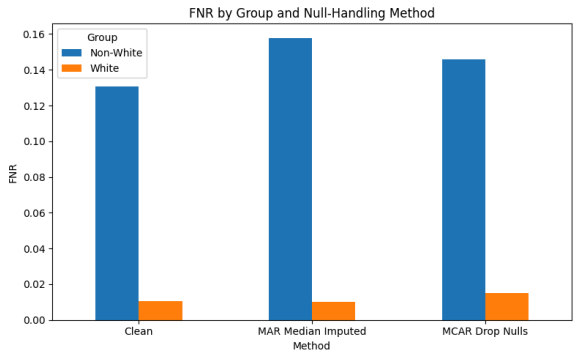


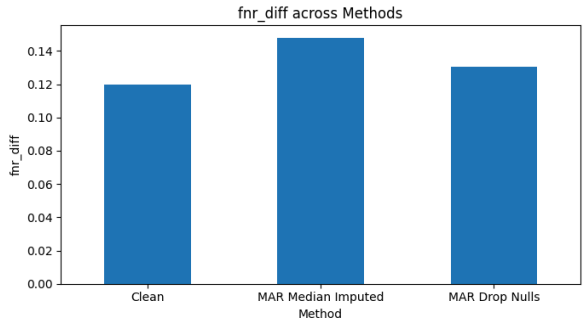
I think the empirical results match the MCAR assumption. There is no strong evidence of systematic bias introduced by either method.

(c) Under the MAR assumption, missingness depends on observed features (in this case, race), meaning that certain groups (e.g., Non-White students) are more likely to have missing values. This introduces systematic bias into the dataset. We expect both deletion and median imputation to affect fairness more significantly than under MCAR, since missingness is no longer random.

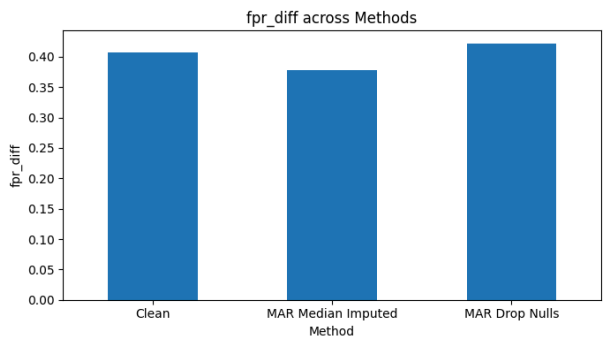
From the graphs we can see that the fnr_diff is higher than under the MCAR assumption, meaning that non-white students are more likely to be mistakenly predicted as not passing the bar. The selection_rate_diff is also slightly higher than the previous result.

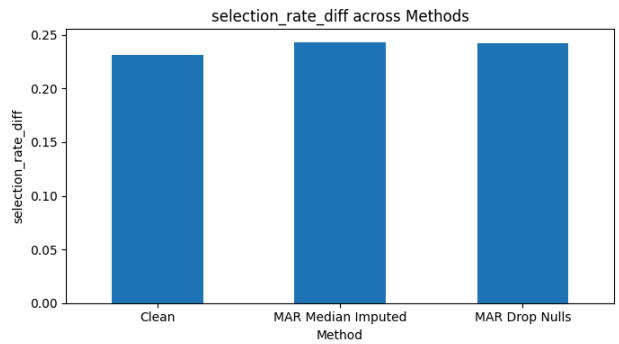
The FPR difference is similar compared to the result of the MCAR method, and it is the highest when deletion is applied.





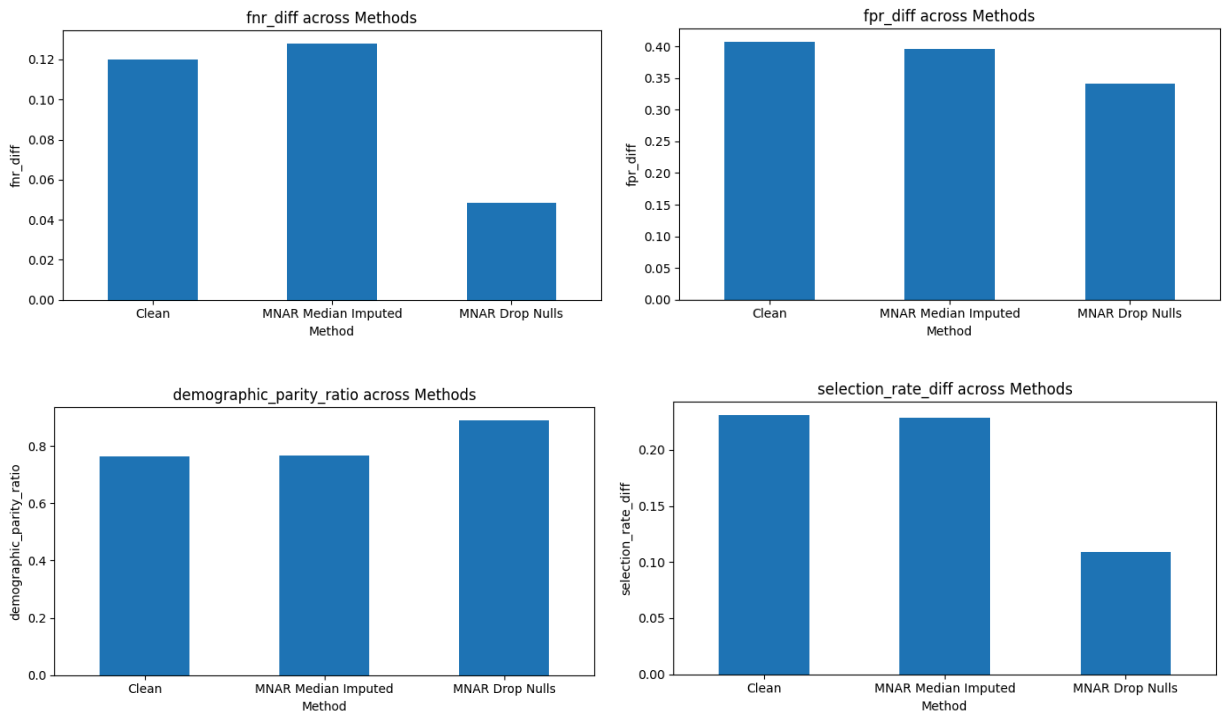
I think the result matches my expectation. Fairness disparities become larger compared to the clean-data baseline, confirming that missingness correlated with race introduces systematic bias. Since there are more missing values in non-white group, deleting them removes more samples from this group, and the model might learn from the sample containing mostly white students and become biased toward white group. Also, although imputing the missing values retain all samples, replacing them all with medians makes the dataset not representing the true distribution.



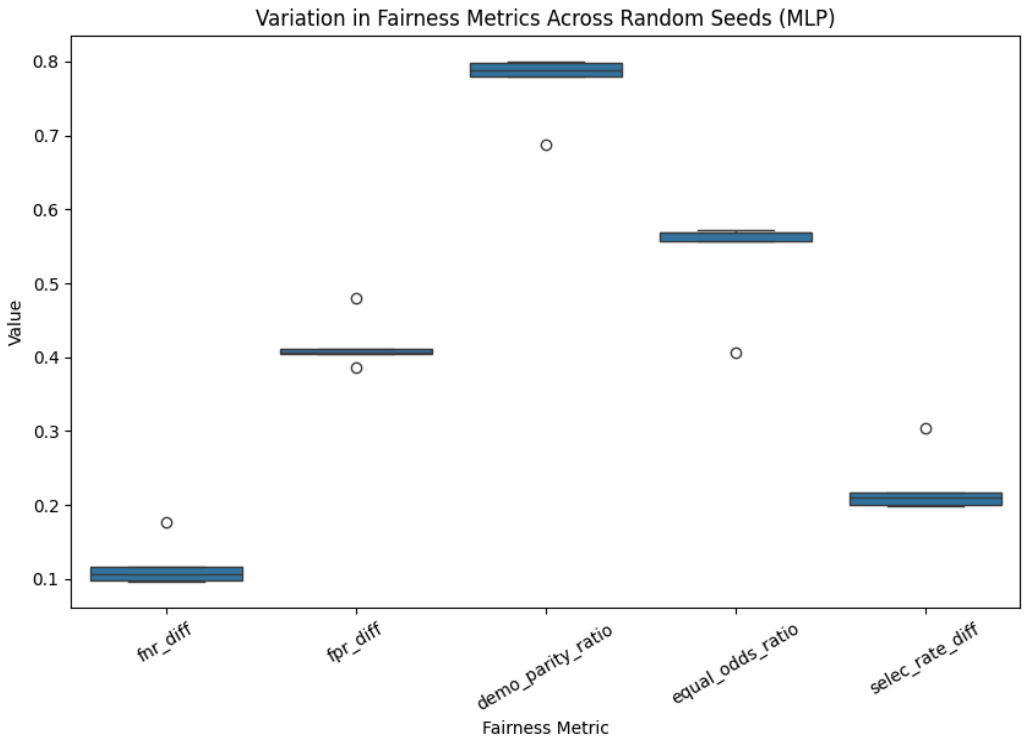


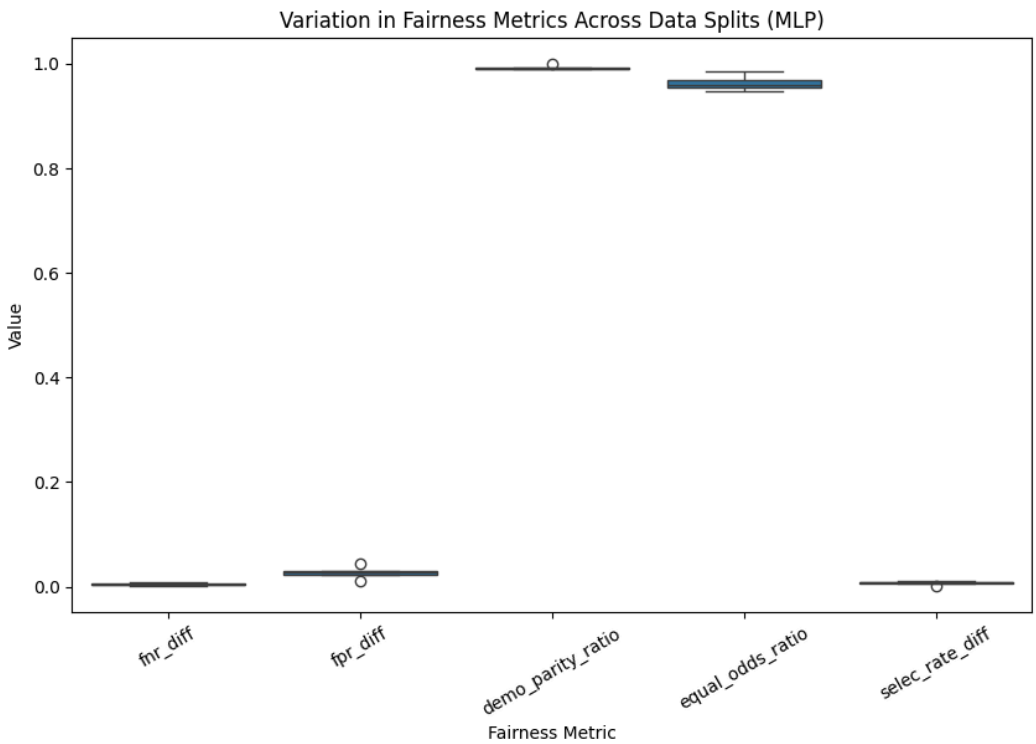
(d) From the result of the MNAR mechanism, it seems like the model performs similarly to the original clean model, and the one using deletion method is even better in fairness metrics (lower fpr_diff and fnr_diff, as well as higher demographic_parity_ratio, etc). This is not the same as my expectation. I expected the model to be more unfair. However, I think the result might be misleading. By removing low-performing individuals, the dataset may become easier to predict. My hypothesis is that: low-performing individuals are where racial disparity is most obvious when a model is biased. By

deleting many of these individuals, the model can look fairer, but I'm not sure whether this is another bias or true fairness.



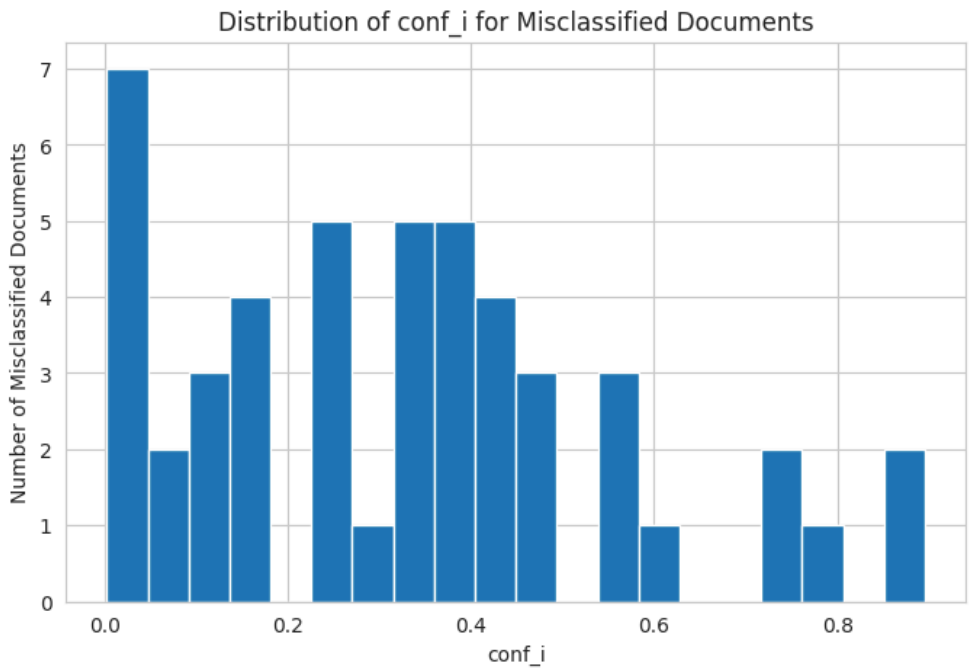
(e) Changing the random state and train-test split leads to noticeable variation in fairness metrics. Even when the model architecture and dataset remain the same, small changes in randomness can significantly alter fairness metrics. Varying the random seeds typically creates more variance than changing the train-test splits. The impact is not uniform across fairness metrics, but we can not see a certain pattern from the graph. A practitioner can do d-hacking by selecting the model with lowest fnr_diff and fpr_diff, or with higher demographic parity ratio or equal_odds_ratio, depending on which metric they value at most.


Variation in Fairness Metrics Across Random Seeds (MLP)
Value
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
fnr_diff
fpr_diff
demo_parity_ratio
equal_odds_ratio
selec_rate_diff
Fairness Metric

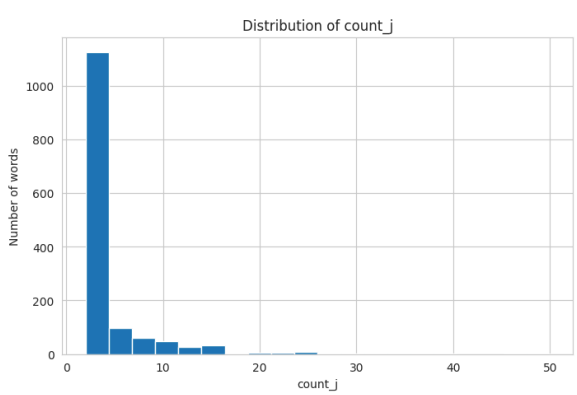

Variation in Fairness Metrics Across Data Splits (MLP)
Value
1.0
0.8
0.6
0.4
0.2
0.0
fnr_diff
fpr_diff
demo_parity_ratio
equal_odds_ratio
selec_rate_diff
Fairness Metric

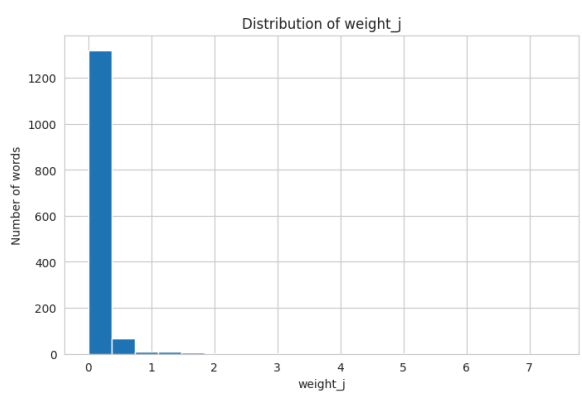
Problem 3

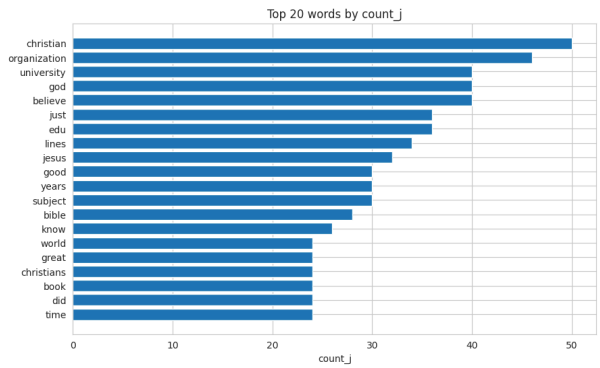
(c) My model has an accuracy of about 0.933. It classified 48 documents wrongly. The plot demonstrates that the distribution of conf_i for misclassified documents is slightly skewed to the right. Many documents have a pretty low confidence.

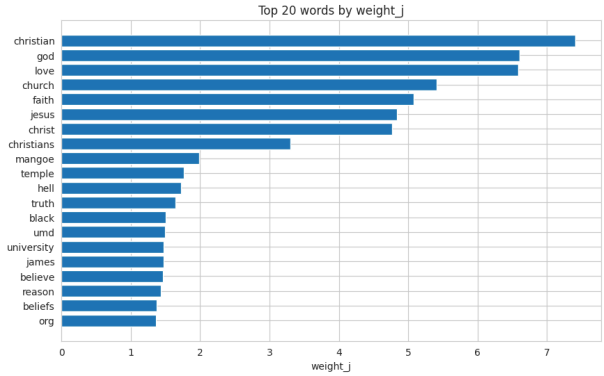


The distribution of count_j and weight_j are heavily skewed to the right. Most words contributed to a small number of mistakes, but some contributed to more than 30, such as “christian”, organization”, and “God”. These words also tend to have a high weight. I believe these words appear both in Atheism and Christian texts.









(d) I deleted 20 words that have the highest weight. I believe they are the main cause of misclassifying a document. After feature selection, the model accuracy improved by 0.014. It corrected 15 documents that were previously misclassified. Although this is not a big improvement, it is meaningful because it corrects some documents that repeatedly use words like “God” but actually demonstrates atheism ideas. From the document I selected, we can see that although this (part of) text uses “God” many times, it is actually refuting some ideas of Christianity.

```
Number of removed words: 20
Some removed words: ['christian', 'god', 'love', 'church', 'faith', 'jesus', 'christ', 'christians', 'mangoe', 'temple', 'hell', 'truth',
Original vocab size: 10000
Filtered vocab size: 9980
Accuracy before feature selection: 0.9330543933054394
Accuracy after feature selection: 0.9470013947001394
Accuracy improvement: 0.013947001394700065
```

Number of corrected examples: 15
Example index: 123
True label: atheism
Prediction BEFORE: christian
Prediction AFTER: atheism

--- Document ---

From: kempmp@phoenix.oulu.fi (Petri Pihko)
Subject: Re: Christian Morality is
Organization: University of Oulu, Finland
X-Newsreader: TIN [version 1.1 PL6]
Lines: 79

Dan Schaertel,,, (dps@nasa.kodak.com) wrote:

Since this is alt.atheism, I hope you don't mind if we strongly disagree...

: The fact is God could cause you to believe anything He wants you to.
: But think about it for a minute. Would you rather have someone love
: you because you made them love you, or because they wanted to
: love you. The responsibility is on you to love God and take a step toward
: Him. He promises to be there for you, but you have to look for yourself.